

UNIT 1

INTRODUCTION TO RAPID PROTOTYPING

One of the important steps prior to the production of a functional product is building of a physical prototype. Prototype is a working model created in order to test various aspects of a design, illustrate ideas or features and gather early user feed-back. Traditional prototyping is typically done in a machine shop where most of parts are machined on lathes and milling machines. This is a subtractive process, beginning with a solid piece of stock and the machinist carefully removes the material until the desired geometry is achieved. For complex part geometries, this is an exhaustive, time consuming, and expensive process. A host of new shaping techniques, usually put under the title ***Rapid Prototyping***, are being developed as an alternative to subtractive processes. These methods are unique in that they add and bond materials in layers to form objects. These systems are also known by the names ***additive fabrication, three dimensional printing, solid freeform fabrication (SFF), layered manufacturing*** etc. These additive technologies offer significant advantages in many applications compared to classical subtractive fabrication methods like formation of an object with any geometric complexity or intricacy without the need for elaborate machine setup or final assembly in very short time. This has resulted in their wide use by engineers as a way to reduce time to market in manufacturing, to better understand and communicate product designs, and to make rapid tooling to manufacture those products. Surgeons, architects, artists and individuals from many other disciplines also routinely use this technology.

- **Prototype:** It is a model fabricated to prove out a concept or an idea.

- Solid Modelling: It's a branch of CAD that produces 2D or 3D objects in an electronic format.
- **Definition: Rapid prototyping is basically a additive manufacturing process used to quickly fabricate a model of a part using 3-D CAM data.**
- **It can also be defined as layer by layer fabrication of 3D physical models directly from CAD.**

Need for the compression in the product development

- To increase effective communication.
- To decrease development time.
- To decrease costly mistakes.
- To minimize sustaining engineering changes.
- To extend product life time by adding necessary features & eliminating redundant features early in the design.

Trends in manufacturing industries emphasize the following:

- Increasing the no of variants of products.
- Increase in product complexity.
- Decrease in product lifetime before obsolescence.
- Decrease in delivery time.
- Product development by Rapid prototyping by enabling better communication.

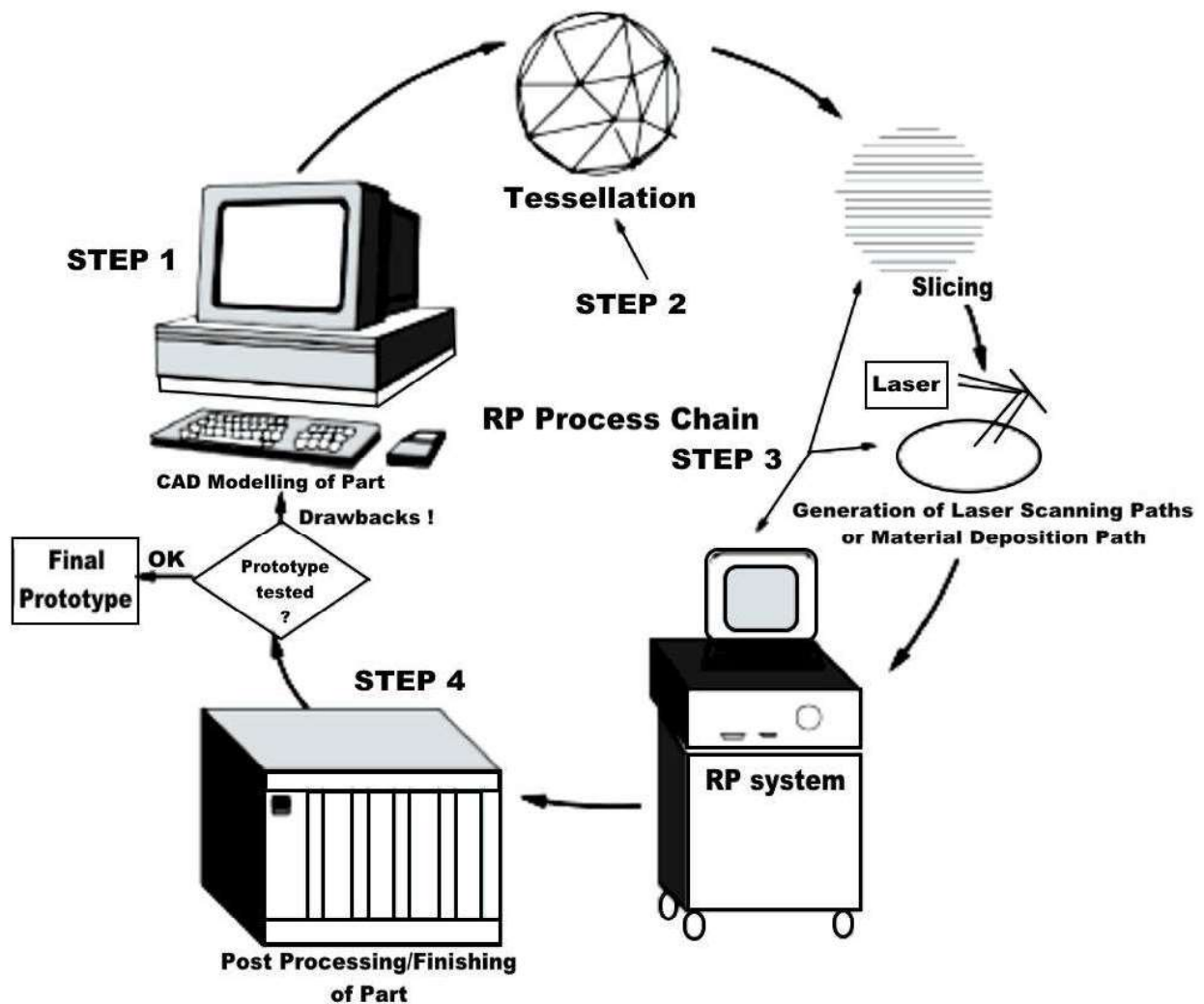
Conventional Machining:

- Its not suitable for complex shapes because they are difficult to machine.
- Time consuming
- Very costly
- Tedious or very laborious.
- Skilled operator is required.
- Accuracy will be less.
- Increased product development time.

- ❖ **Pre-processing:-** CAD model slicing & setting algorithms applied for various RP systems.
- ❖ **Post-processing:-** Cleaning operations required to finish a part after removing it from RP machine.
- ❖ **Materials for Rapid Prototyping:-** Paper, Wax, Plastics, Resins, Metallic powders.

Methodology of Rapid Prototyping:

RP in its basic form can be described as the production of three dimensional (3D) parts from computer aided design (CAD) data in a decreased time scale. The basic methodology of all RP process can be summarized as shown in following figure.



Rapid prototyping process chain

- Construct a CAD model.
- Convert it to STL format.
- RP machine processes .STL file by creating sliced layers of model.
- First layer of model is created.
- Model is then lowered by thickness of next layer.
- Process is repeated until completion of model

- The model & any supports are removed.
- Surface of the model is then finished and cleaned.

(1) Development of a CAD model

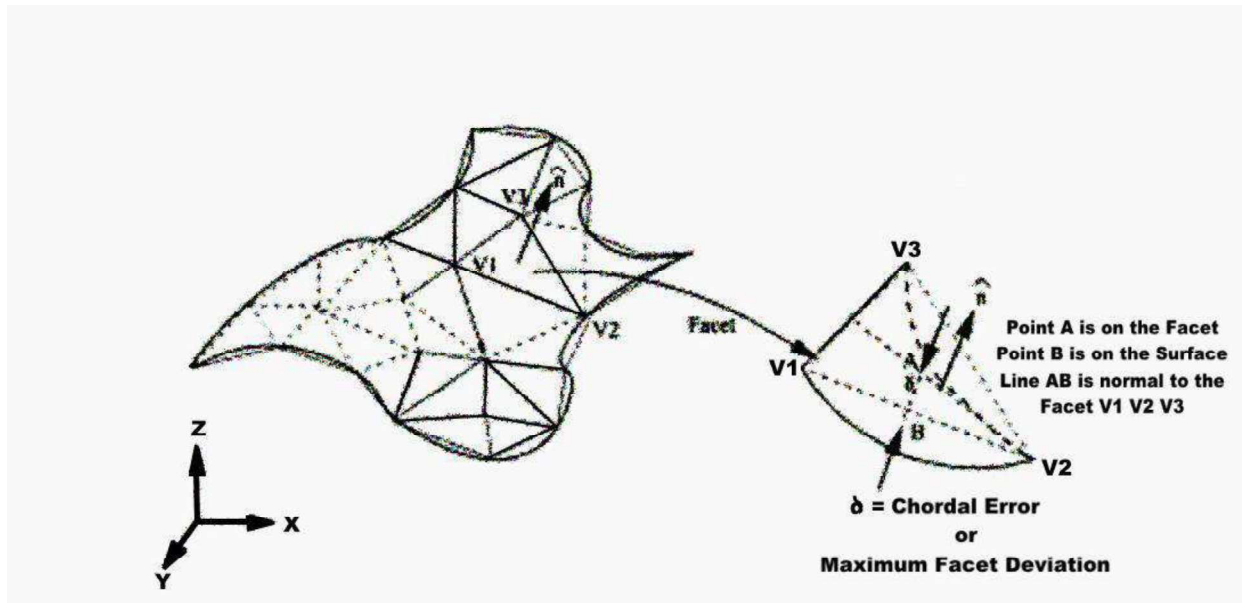
The process begins with the generation CAD model of the desired object which can be done by one of the following ways;

- Conversion of an existing two dimensional (2D) drawing
- Importing scanned point data into a CAD package
- Creating a new part in CAD in various solid modeling packages
- Altering an existing CAD model

RP has traditionally been associated with solid rather than surface modelling but the more recent trends for organic shapes in product design is increasing the need for free flowing surfaces generated better in surface modelling.

(2) Generation of Standard triangulation language (STL) file

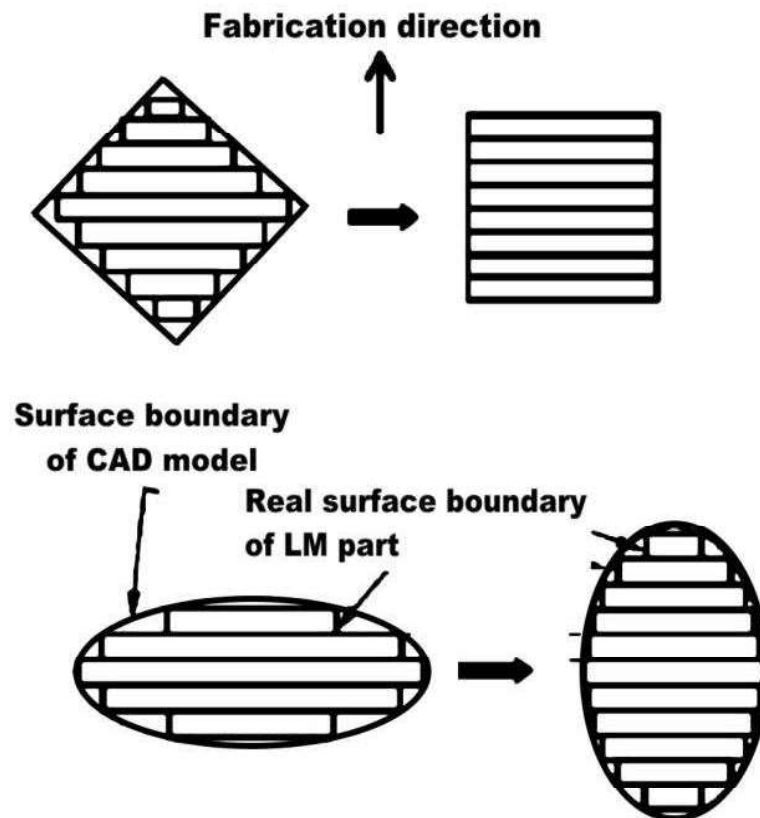
The developed 3D CAD model is tessellated and converted into STL files that are required for RP processes. Tessellation is piecewise approximation of surfaces of 3D CAD model using series of triangles. Size of triangles depends on the chordal error or maximum facet deviation. For better approximation of surface and smaller chordal error, small size triangle are used which increase the STL file size. This tessellated CAD data generally carry defects like gaps, overlaps, degenerate facets etc which may necessitate the repair software. These defects are shown in figure below. The STL file connects the surface of the model in an array of triangles and consists of the X, Y and Z coordinates of the three vertices of each surface triangle, as well as an index that describes the orientation of the surface normal.



Tessellation defects

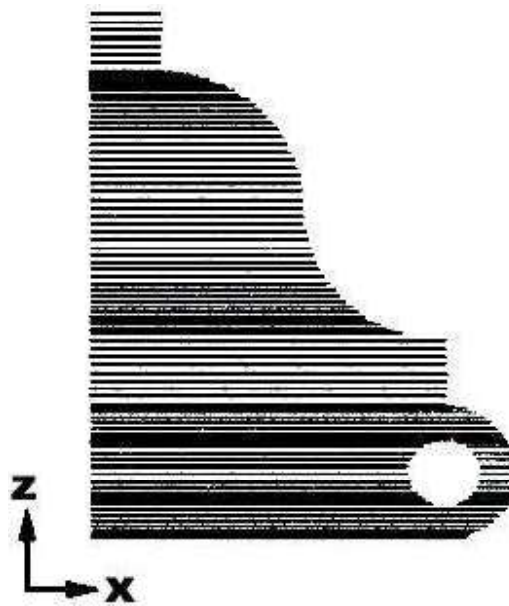
(3) Slicing the STL file

Slicing is defined as the creating contours of sections of the geometry at various heights in the multiples of layer thickness. Once the STL file has been generated from the original CAD data the next step is to slice the object to create a slice file (SLI). This necessitates the decision regarding part deposition orientation and then the tessellated model is sliced. Part orientation will be showing considerable effect on the surface as shown in the figures.



Effect of Part deposition Orientation

The thickness of slices is governed by layer thickness that the machine will be building in, the thicker the layer the larger the steps on the surface of the model when it has been built. After the STL file has been sliced to create the SLI files they are merged into a final build file. This information is saved in standard formats like SLC or CLI (Common Layer Interface) etc.



(4) Support Structures

As the parts are going to be built in layers, and there may be areas that could float away or of overhang which could distort. Therefore, some processes require a base and support structures to be added to the file which are built as part of the model and later removed.

(5) Manufacturing

As discussed previously, the RP process is additive i.e. it builds the parts up in layers of material from the bottom. Each layer is automatically bonded to the layer below and the process is repeated until the part is built. This process of bonding is undertaken in different ways for the various materials that are being used but includes the use of Ultraviolet (UV) lasers, Carbon Dioxide (CO_2) lasers, heat sensitive glues and melting the material itself etc.

(6) Post processing

The parts are removed from the machine and post processing operations are performed sometimes to add extra strength to the part by filling process voids or finish the curing of a part or to hand finish the parts to the desired level. The level of post processing will depend greatly on the final requirements of the parts produced, for example, metal tooling for injection molding will require extensive finishing to eject the parts but a prototype part manufactured to see if it will physically fit in a space will require little or no post processing.

History of RP system

- It started in 1980's
- First technique is Stereo lithography (SLA)
- It was developed by 3D systems of Valencia in California, USA in 1986.
- Fused deposition modeling (FDM) developed by stratasys company in 1988.
- Laminated object manufacturing (LOM) developed by Helisis (USA).
- Solid ground Curing developed by Cubitol corporation of Israel.
- Selective laser sintering developed by DTM of Austin, Texas (USA) in 1989.
- Sanders Model maker developed by Wilton incorporation USA in 1990.
- Multi Jet Modeling by 3D systems.
- 3-D Printing by Solygen incorporation, MIT, USA.